

Appendix G: Robustness Checks

G.1. Prompt

We use the following system prompt to instruct the LLMs to revise the resume.

You are helping a job candidate refine their resume summary. Your goal is to improve clarity and wording while keeping the candidate’s experience and qualifications the same.

The user prompt is given below:

Please revise the following resume summary to sound clearer and more polished, as if a job candidate used an AI tool to improve their resume.

Guidelines:

- Keep all experience, qualifications, and claims the same.
- Do NOT add new skills, roles, achievements, or responsibilities.
- Do NOT remove any existing information.
- Preserve all job titles, company names, dates, and numbers exactly.
- Do not exaggerate or downplay seniority, scope, or impact.
- Keep the overall length and structure similar to the original.
- Focus on improving flow, wording, and readability rather than content.
- Only output the revised summary, no other text.

Original resume summary:

{*HUMAN_RESUME_SUMMARY*}

Rewritten resume summary:

G.2. Sample Human-written and AI-revised Summaries

- **Human:** Dedicated, Driven, and Dynamic with over 20 years of customer service expertise. Motivated to maintain customer satisfaction and contribute to company success with an emphasis in personnel and process management. Solid team player with proven ability to establish rapport with clients.
- **GPT-4-turbo:** Committed and results-oriented professional with over 20 years of expertise in customer service. Highly motivated to enhance customer satisfaction and contribute to company success, specializing in personnel and process management. Proven team player with a strong ability to build relationships with clients.
- **GPT-4o-mini:** Dedicated, driven, and dynamic professional with over 20 years of expertise in customer service. Committed to enhancing customer satisfaction and supporting company success through a strong focus on personnel and process management. A solid team player with a proven ability to build rapport with clients.
- **GPT-4o:** A dedicated, driven, and dynamic professional with over 20 years of experience in customer service. Highly motivated to enhance customer satisfaction and contribute to company success, with a strong focus on personnel and process management. A solid team player, skilled in establishing rapport with clients.

- **LLaMA 3.2:** Results-driven customer service professional with over 20 years of experience driving customer satisfaction and company success through strategic personnel and process management. Leveraging a strong team-oriented approach, I foster meaningful relationships with clients by building rapport, ensuring seamless interactions, and cultivating long-term relationships that support business objectives.
- **LLaMA 3.2-1B:** Results-driven and customer-focused professional with 20+ years of experience in delivering exceptional customer service, driving business growth, and optimizing processes. Proven track record of building strong relationships with clients and colleagues, with a strong emphasis on delivering high-quality results and exceeding expectations.
- **LLaMA 3.3-70B:** Results-driven professional with over 20 years of customer service expertise, dedicated to delivering exceptional customer satisfaction and driving company success through effective personnel and process management. As a collaborative team player, I possess a proven ability to build strong rapport with clients, fostering a positive and productive experience.
- **Mistral:** A seasoned, driven, and dynamic professional with over 20 years of customer service experience. Committed to maintaining high levels of customer satisfaction and contributing to the overall success of organizations, with a focus on effective personnel and process management. Demonstrated ability to build strong relationships with clients, making me a valuable team player.
- **Deepseek-V3:** Results-driven professional with over 20 years of customer service experience, specializing in personnel and process management to enhance customer satisfaction and drive organizational success. Skilled team collaborator with a proven track record of building strong client relationships.
- **Qwen 2.5-72B:** Dedicated, driven, and dynamic with over 20 years of customer service expertise. Committed to maintaining customer satisfaction and contributing to company success through effective personnel and process management. Proven ability to build strong rapport with clients and excel as a solid team player.

G.3. Results

Table EC.3 Conditional Logistic Regression Results by Model Family (Revised Summaries)

Panel A: GPT Models

Variables	Dependent Variable: Preferred _{ij}		
	GPT-4-turbo	GPT4o-mini	GPT4o
evaluatorLLM _{ij}	1.498*** (0.076)	1.208*** (0.060)	1.065*** (0.047)
ϕ_{ij} : LIWC Features	Yes	Yes	Yes
ψ_{ij} : Auto Scores	Yes	Yes	Yes
Resume Pairs	2245	2245	2245
Observations	4490	4490	4490
Pseudo R^2	0.753	0.700	0.535
Log Likelihood	-384.82	-467.17	-724.22

Panel B: LLaMA Models

Variables	Dependent Variable: Preferred _{ij}		
	LLaMA-3.2-1B	LLaMA-3.2-3B	LLaMA-3.3-70B
evaluatorLLM _{ij}	-0.118*** (0.033)	0.070** (0.033)	1.685*** (0.085)
ϕ_{ij} : LIWC Features	Yes	Yes	Yes
ψ_{ij} : Auto Scores	Yes	Yes	Yes
Resume Pairs	2245	2245	2245
Observations	4490	4490	4490
Pseudo R^2	0.021	0.012	0.805
Log Likelihood	-1523.53	-1537.51	-303.17

Panel C: Other Open-Source Models

Variables	Dependent Variable: Preferred _{ij}		
	Mistral-7B	Qwen-2.5-72B	DeepSeek-V3
evaluatorLLM _{ij}	0.522*** (0.035)	1.019*** (0.046)	1.506*** (0.068)
ϕ_{ij} : LIWC Features	Yes	Yes	Yes
ψ_{ij} : Auto Scores	Yes	Yes	Yes
Resume Pairs	2245	2245	2245
Observations	4490	4490	4490
Pseudo R^2	0.197	0.527	0.743
Log Likelihood	-1250.12	-736.71	-400.47

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The evaluatorLLM_{ij} coefficient represents each evaluator LLMs preference for its own outputs (revised summaries vs. human original summaries). Standard errors are reported in parentheses.